

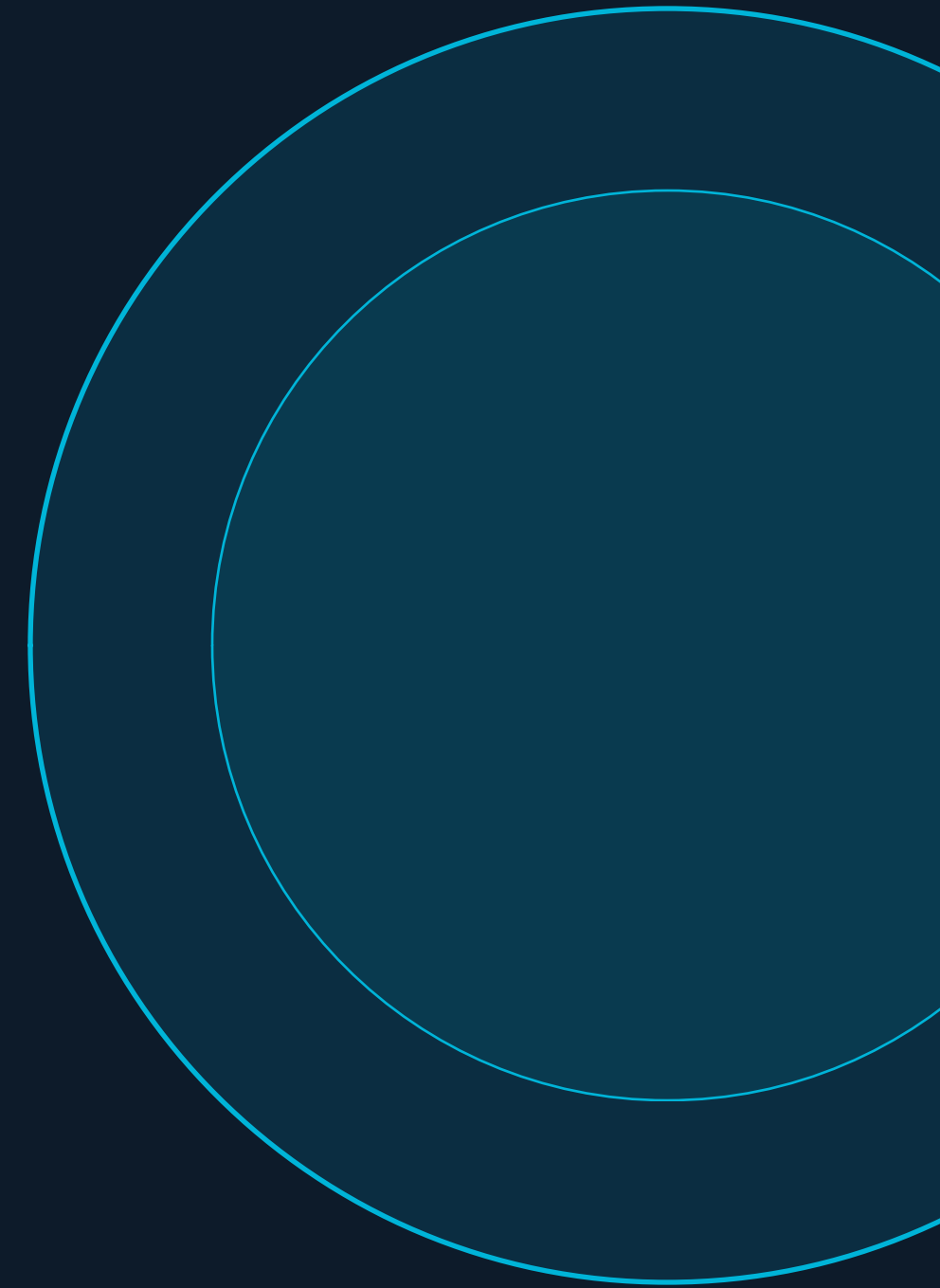
EE655 Project Presentation

Human Body Matting

Using U-Net with Combined Dice-BCE Loss

Dice Score: 0.98 · Mean IoU: 0.93 · Streamlit Deployed

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Problem Statement

① Input & Output

Input: RGB portrait image (any resolution)

Output: Binary foreground mask

Use Case: Virtual backgrounds, portrait mode, green-screen replacement

② Constraints

Lightweight: must run on single GPU (no TPU/cloud)

Open-source: no commercial closed-source tools

Fine edges: hair, arms, shoulders must not blur

Class imbalance: background >> foreground pixels

End-to-End Pipeline



Literature Review

Simple BCE Loss

Standard binary cross-entropy ignores class imbalance — background pixels dominate, causing blurry edges.

No Augmentation

Baseline models train on raw data only, leading to poor generalization across varied poses and lighting.

Fixed Threshold (0.10)

Hardcoded threshold inflates recall at the cost of precision, creating noisy mask boundaries.

Commercial Tools

Closed-source matting tools (Adobe, Zoom) perform well but require expensive hardware and licenses.

How Our Work Differs

- Combined Dice+BCE loss → optimizes shape overlap, not just pixel accuracy
- Spatially-consistent augmentation on image+mask simultaneously
- Empirical threshold sweep → $\tau^* = 0.30$ (vs. hardcoded 0.10 / 0.50)

Proposed Method — Key Innovations

1 Combined Dice+BCE Loss

- Why: Standard BCE ignores foreground/background imbalance.
- $L = L_{\text{Dice}} + L_{\text{BCE}}$
- Dice term: $1 - (2 \cdot \sum y \hat{y} + \epsilon) / (\sum y + \sum \hat{y} + \epsilon)$
- Dice coefficient = $2TP / (2TP + FP + FN)$
- Both tensors flattened to 1D before arithmetic — eliminates `InvalidArgumentError`.
- $\epsilon = 1e-7$ for numerical stability.

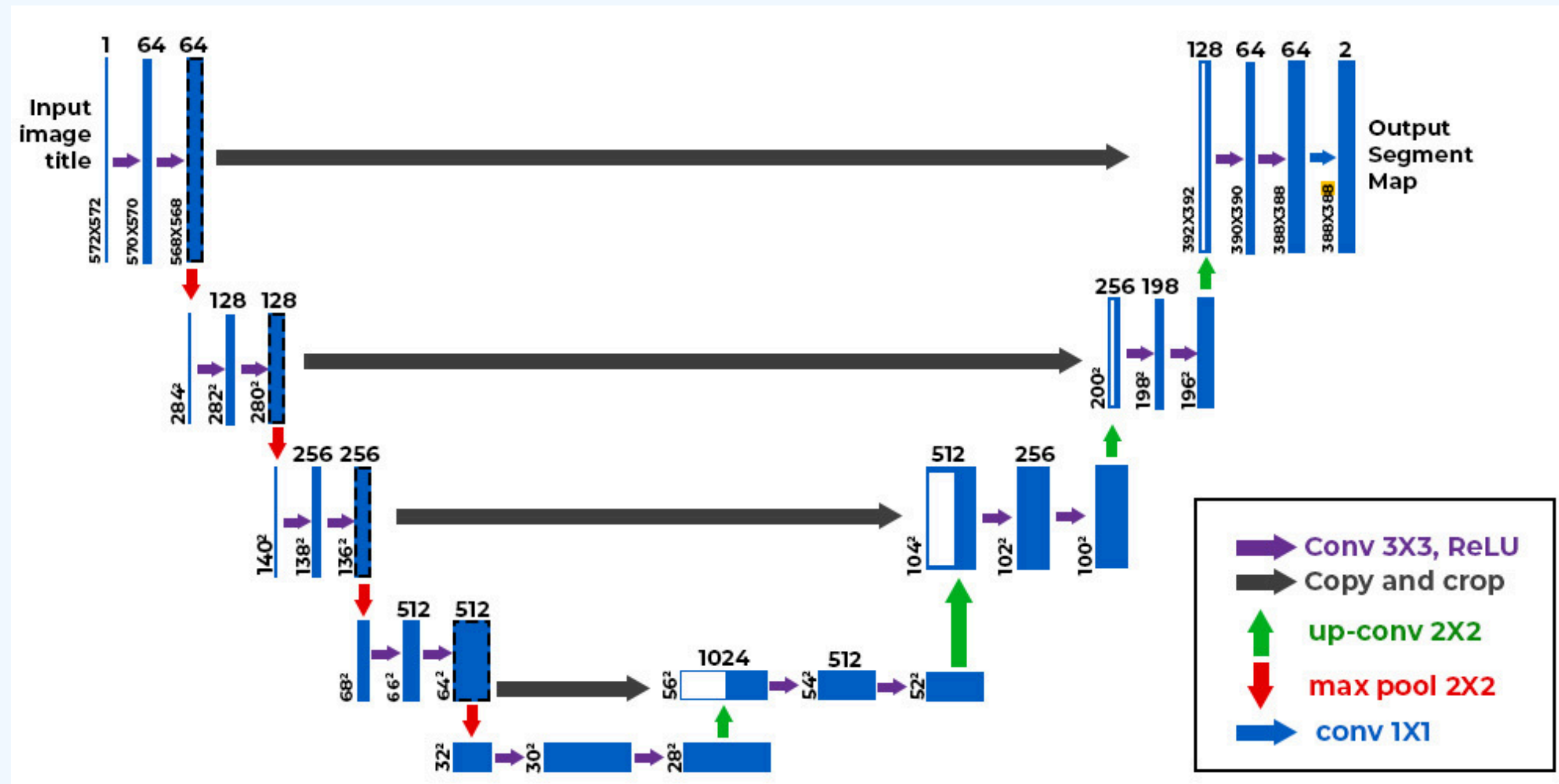
2 Data Augmentation

- Why: Baseline had zero augmentation; model overfit to training poses.
- Horizontal flip ($p=0.5$) applied to image AND mask simultaneously.
- Brightness jitter ± 0.15 on image only.
- Contrast jitter $\in [0.8, 1.2]$ on image only.
- Prevents label misalignment between image and mask.

3 Optimal Threshold (τ^*)

- Why: Hardcoded 0.10 inflates recall, producing noisy masks.
- Sweep 16 values from 0.10 $\rightarrow$ 0.85 (step 0.05).
- Evaluated on 50-image validation sample.
- $\tau^* = 0.30 \rightarrow$ peak Dice = 0.9733.
- Replaces guess with empirical evidence.

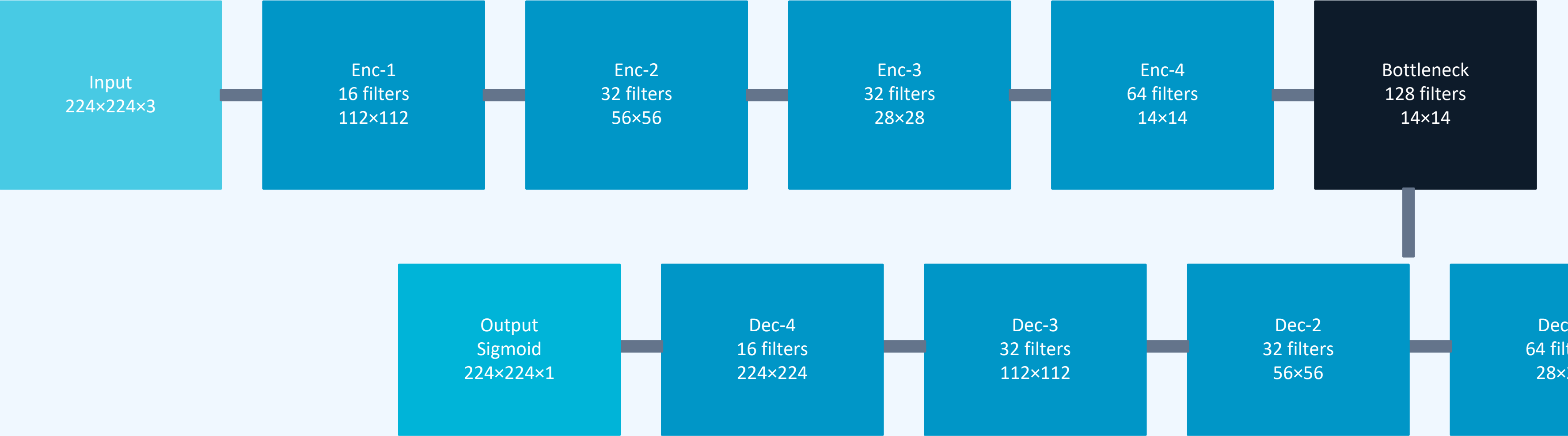
Proposed Method — U-Net Architecture



Total Parameters: 594,577
Trainable: 592,913

Activations: SELU · Regularization: Dropout + BatchNorm
Final Layer: 1x1 Conv2D + Sigmoid

Proposed Method — U-Net Architecture



Skip connections (concatenate encoder feature maps with decoder)

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Experiments — Dataset & Setup

Dataset

AISeam.com Matting Human Dataset

Source: Kaggle

- 34,427 image-mask pairs
- Professional alpha matte annotations
- Diverse poses, clothing, lighting
- Train: 1,000 sampled (with augmentation)
- Validation: 500 images

Hyperparameters

Framework	TensorFlow 2.6.4
Input Size	224 × 224 × 3
Optimizer	Adam, lr = 1e-3
Batch Size	16
Max Epochs	100 (stopped at 20)
Early Stop	patience = 7, val_loss
Hardware	NVIDIA Tesla P100

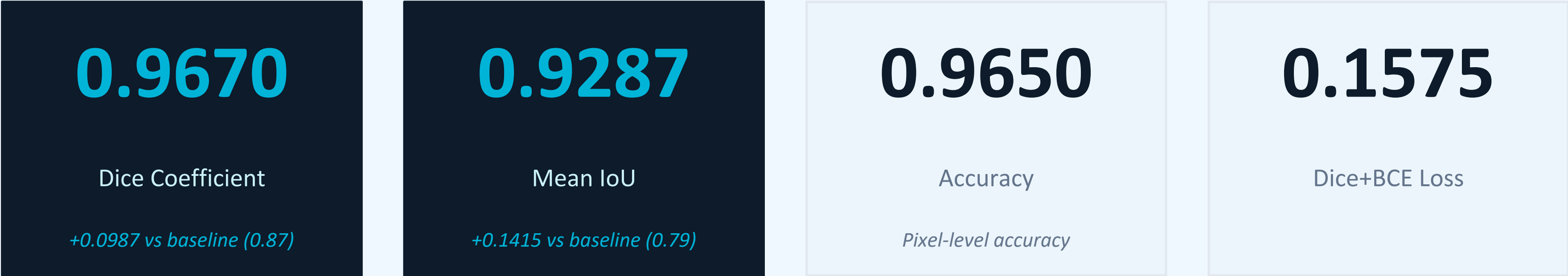
Training Summary

Epoch 1: train_loss=0.0784, val_loss=0.1896

Epoch 13 (BEST): train_loss=0.0535, val_loss=0.1662★

Epoch 20: Early stopping triggered → best weights restored

Results — Quantitative Evaluation

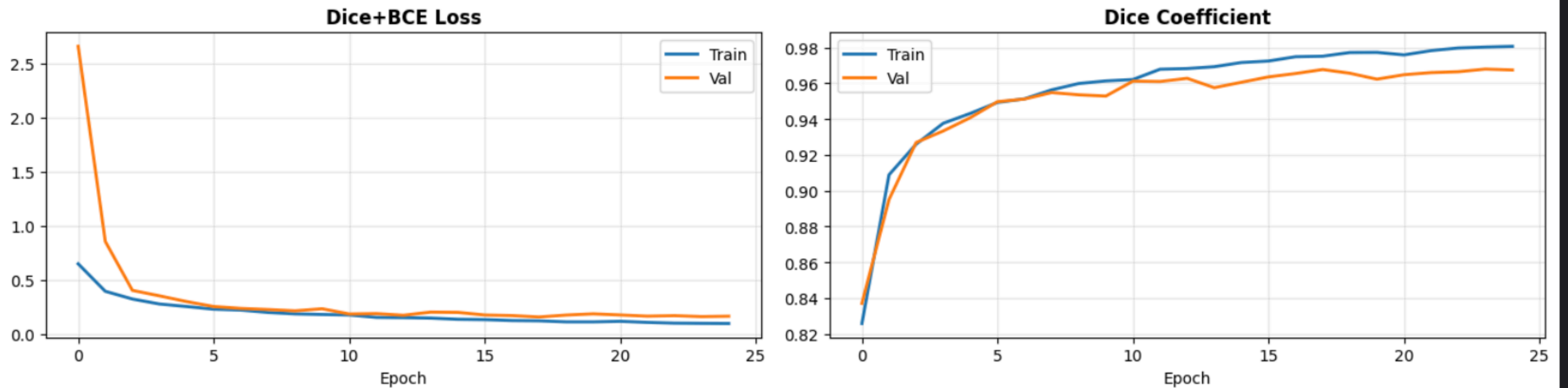


Comparison: Our Model vs Baseline (BCE-only)

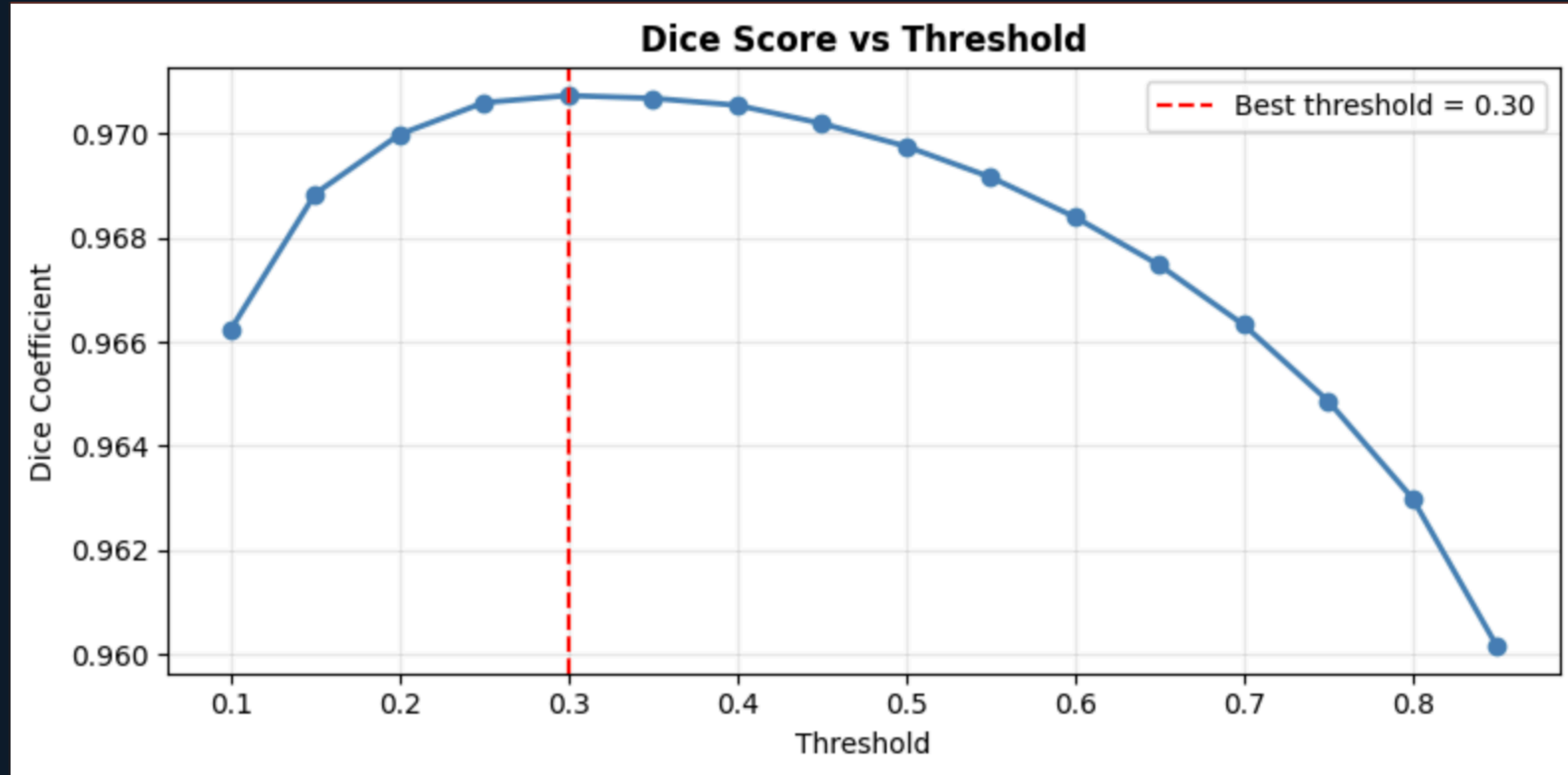
Metric	Baseline (BCE)	Our Model	Improvement
Dice Score	0.87	0.9670	+11.3%
Mean IoU	0.79	0.9287	+17.9%
Loss Function	BCE only	Dice + BCE	Combined
Threshold	0.10 (hardcoded)	0.30 (swept)	Empirical

Results — Training Curves

Training History



Results — Threshold Analysis



Optimal threshold: 0.30 | Best Dice: 0.9707

Threshold Sweep: 16 thresholds [0.10 → 0.85]. Optimal $\tau^* = 0.30 \rightarrow \text{Dice} = 0.9707$. Baseline value 0.10 sits on the declining tail.

Discussion & Limitations

✓ What Worked

- Combined loss: optimizes overlap not just pixels — biggest single gain
- Augmentation: improved generalization without extra labeled data
- Threshold sweep: only 1 forward pass on 50 images, major precision gain
- Qualitative: hair and shoulder edges well preserved in predictions

✗ Limitations

- Random batch sampling — not all 34k images used; unusual poses may fail
- FixedMeanIoU had to be excluded from training compile — Keras crash with Dice+BCE
- Only 1,000 train images per run; full dataset training would likely help

Future Work: Attention gates in decoder · Transformer bottleneck for fine hair details · Full 34k dataset training · Real-time webcam deployment

Conclusion

Architecture

5-level U-Net (594K params) with SELU, BN, Dropout — lightweight and effective

Loss Function

Dice+BCE jointly optimizes overlap and pixel accuracy; fixes class imbalance

Augmentation

Spatially consistent flip + brightness/contrast jitter; prevents label misalignment

Threshold

Empirical sweep found $\tau^*=0.30$ (Dice=0.9670) vs hardcoded 0.10 baseline

Final Results: Dice 0.87 $\rightarrow$ 0.9670 · IoU 0.79 $\rightarrow$ 0.9287 · Deployed via Streamlit + ngrok

Thank You!